

≡ *Equilux*

Pay, proven equal. No salary revealed.

The first employee-verifiable pay-transparency reporting protocol. A company proves its legally required gender pay gap report was computed from its complete payroll — every salary the payroll provider's own figure, confirmed by the employee it belongs to — while every salary stays in Midnight's private state.

Midnight Buildathon · Wave 2

Compact · TypeScript · Apache 2.0

github.com/AustinChris1/Equilux

The law now demands a number companies cannot prove.

7 June
2027

FIRST REPORTS DUE • 2026 PAY
DATA

250+

EMPLOYEES, ANNUAL • 150+
SAME DATE, TRIENNIAL

≥ 5%

UNJUSTIFIED GAP IN A
CATEGORY → JOINT PAY
ASSESSMENT

Article 9 requires seven indicators. Article 7 gives every employee the right to ask what their category earns. And today every one of those reports is **self-declared**: nobody can verify the numbers without seeing everyone's salary.

Trust me, or prove it.

Today

HR exports payroll to a spreadsheet, computes a gap, publishes a PDF. An employee cannot check they were counted. A works council cannot check nobody was dropped. A regulator can only audit by subpoenaing the whole payroll — the data the law is trying to protect.

With Equilux

The payroll provider — not the employer — commits every payroll row, hashed. Each employee seals their record and proves it matches their row. A zero-knowledge circuit verifies the gaps over the sealed set and publishes only the figures plus a proof that they came from all of it. Verification takes seconds. No salary ever leaves the company.

Equilux does not replace the HRIS. Payroll stays in Personio, DATEV or Workday. Equilux is the **notary on top**: it proves the set was complete and the published figure matches it.

The payroll provider vouches for the figures. A plugin, not an app.

	WAVE 1	WAVE 2
Where each salary comes from	the employer, for its own data	the provider's payroll row , confirmed by the employee
Leaving someone out · inventing someone	blocked only if they enrolled	both blocked — payroll committed before enrollment
Statistics proven	mean gap	mean · median · per worker category
Article 7 category pay	designed	published at k = 3 , small groups suppressed
The public site	a replica of the rules	the compiled contract, in the browser
HRIS integration	—	Personio / DATEV CSV import + party API
Tests on the compiled circuits	9	28

Trust neither side. **Verify both.**

Completeness

The payroll provider commits every payroll row before anyone enrolls; the report must cover exactly those rows. Leaving someone out — even by never enrolling them — is rejected.

Every salary is payroll's

A record counts only if it opens to an unclaimed payroll row. The employer cannot invent a person or change anyone's pay; an employee cannot inflate their own.

No small-group leaks

A category's pay is published only when both groups have at least three people. Smaller groups show headcounts only.

Exact figures

Mean gap, median gap and every category's pay and gap are verified in-circuit — no division, no sort. One basis point off fails.

Plus an **inclusion receipt** for every employee: a Merkle path proving their record is inside the numbers, revealing nothing about anyone.

Payroll commits. The employee confirms. The circuit checks everything.

01 • declareRoster

Commit payroll

The payroll provider commits a hiding hash of every payroll row, in one transaction.

02 • enroll

Seal & bind

The employee seals their record and proves it opens to their payroll row, with the nonce from their payslip.

03 • publishReport

Prove

Every row covered, every record bound and distinct; mean, median and category gaps verified.

04 • checkReceipt

Verify

An employee proves their record is in the tree against the on-chain root.

Private state holds the payroll. Public state holds the proof.

Private state • witnesses

- salary, gender marker, worker category
- employee enrollment secret and payslip nonce
- employer and payroll-provider secret keys
- the full 16-slot payroll witness
- every claimed figure (verified, never trusted)

Public ledger

- commitments — a depth-5 Merkle tree
- payroll-row hashes, nullifiers, bound commitments
- employer + provider keys (sealed), headcount
- `PayReport` : headcounts, mean + median gaps, 4 category rows ($k = 3$)
- the zero-knowledge proof for every transition

Every witness value that reaches the ledger passes an explicit `disclose()`. The compiler refuses anything else.

Fourteen people. Real proofs. One proven report.

4

CIRCUITS · PROVER + VERIFIER KEYS

28

TESTS · COMPILED CIRCUITS VIA COMPACT-RUNTIME

0

SALARIES ON CHAIN · EVER

deployed at `620570fe...6679` · employer + payroll-provider keys (block 27)
provider commits `14 payroll-row hashes` in one transaction (block 31)
14 enrollments, each proven to open its own payroll row (blocks 36-91)
ada claims €65,000, payroll says €60,000 → `rejected`: matches no payroll row
publishReport `proven & finalized` · mean 9.69% · median 1.81% (block 109)
Engineering 13.88% flagged · Sales 3.70% · Operations suppressed · `same as the live site`

No Docker. No install. The real circuits, on the public URL.

EMPLOYERPAYROLL PROVIDEREMPLOYEEREGULATOR

5 · WHAT THE REPORT SHOWS – AND NOTHING ELSE

MEAN GAP
9.69%
favors men

MEDIAN GAP
1.81%
favors men

HEADCOUNT
14
7 women · 7 men

WORKER CATEGORY	W / M	MEAN PAY W · M	GAP
Engineering	3 / 3	€62,000 · €72,000	13.88% ≥ 5% · assess
Sales	3 / 3	€52,000 · €54,000	3.70% < 5%
Operations	1 / 1	suppressed – fewer than 3 in a group, so no one's pay can be inferred	

A category at ≥ 5% triggers an Article 10 joint pay assessment only if the gap is also unjustified by objective, gender-neutral criteria – a human judgement the circuit does not make. Salaries on chain: 0.

Honest about what a proof can prove.

IT DOES NOT CLAIM	WHAT IS TRUE INSTEAD
Salaries hidden from HR	HR still holds payroll. Hidden from the public, colleagues, the regulator and Equilux — and the employer can no longer supply its own figures.
An honest payroll by itself	A provider colluding with the employer could commit false rows; each employee sees their own row when enrolling. A works-council co-signature is next.
Judging what is justified	Category gaps $\geq 5\%$ are proven and flagged. Whether a gap is justified is a human judgement.
The full Article 9 return	Three of seven indicators: mean, median, category gap. Variable pay and quartiles follow.
Production scale · public testnet	16 records, 4 categories. Local Midnight network — preprod waits on a wallet-SDK fix.
A model of identity	Binary gender markers, as the Directive's reporting annex is written today.

Built in the open, wave by wave.

Wave 1 · delivered

Prove it works

Contract with private state · proven end to end on a Midnight network · employer / employee / regulator workspace · honest scope.

Wave 2 · this wave

The product

Payroll provider as a third party · every salary bound to its payroll row · median and category gaps · $k = 3$ category pay · the compiled contract in the browser · Personio / DATEV import.

Wave 3

Fundable

Public testnet + Lace writes · variable pay and quartiles (all of Article 9) · filing pack with proof hash + verify URL · works-council co-signature · one named pilot.

The notary for Directive 2023/970.

Buyers

Works councils that don't trust the numbers. GDPR-sensitive employers. Regulators who want completeness without holding payroll.

A plugin, not a rival

Workday, Personio, DATEV compute pay. Equilux imports their export, the provider commits its rows, and the report is proven — next to them, not instead of them.

Why Midnight

A forced legal deadline plus a problem only dual-ledger selective disclosure solves honestly. Incumbents can copy a dashboard; they cannot copy the proof.

